K.R. MANGALAM UNIVERSITY

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University Fleet Optimization System

Pre-Final Year Project Synopsis
Submitted by

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Project Overview

The Problem:
Managing university transportation across a massive catchment area(Sohna, Gurugram, Faridabad, Palwal, Noida, Bhiwadi) is manually intensive.
Current fixed routes lead to inefficiencies, high fuel costs, and uneven student loads.

The Solution:
A "Zero-Operational Cost" fleet management system built entirely on open-source technologies.
Replaces expensive commercial APIs (Google Maps) with self-hosted routing engines.

The Goal:
To automate route planning using advanced Multi-Depot Capacitated Vehicle Routing Problem with Time Windows (MD-CVRPTW). Our goal is to minimize total fleet mileage while strictly enforcing bus capacities and campus arrival deadlines.

Specific Objectives

Zero-Cost Architecture:
To achieve professional routing capabilities using self-hosted OSRM and OpenStreetMap data, eliminating recurring API fees.

Regional Optimization:
To effectively model the specific road networks of Haryana and Uttar Pradesh (Extended NCR) for accurate travel time predictions.

Algorithmic Efficiency:
To implement the Vehicle Routing Problem (VRP) using Google OR-Tools to mathematically minimize total fleet mileage.

Constraint Enforcement:
To strictly enforce service rules: Maximum 2-hour ride time and Bus Capacity limits (e.g., 50 seats).

Split-Delivery Logic:
To dynamically split high-demand stops across multiple buses to achieve 100% student coverage without exceeding individual vehicle capacities.

Key Features

No Recurring Costs:
Runs entirely on local university infrastructure or free-tier cloud services.

Physics Aware Routing:
Custom OSRM Lua profile calibrated for “Bus” physics-enforcing strict 40-second U-turn penalties and capping speeds to realistic urban traffic limits.

Strategic Comparison Mode:
Allows admins to toggle between "Strict" plans (visit every stop) and "Suggested" plans (merge stops for efficiency).

High-Performance Dashboard:
A Next.js-based admin panel utilizing PMTiles and Leaflet to visualize thousands of route coordinates seamlessly without browser lag.

Progress Summary

System Architecture (Completed):
Successfully decoupled Next.js frontend and FastAPI backend, communicating securely via REST.

Routing Engine (Live):
Open Source Routing Machine (OSRM) fully self-hosted via docker with custom Extended NCR map data.

Algorithmic Optimization (Validated):
Successfully modeled the MD-CVRPTW, processing datasets of 3000+ students and 100+ buses in minutes.

DevOps & CI/CD (Deployed):
Automated deployment pipeline established using Github Actions, Docker, and Nginx reverse proxies.

Detailed Progress – Development Phase

Phase 1 – Data & Physics:
Engineered a custom Lua profile for OSRM to ensure mathematically generated routes are actually drivable by heavy 12-meter vehicles.

Phase 2 – Algorithmic Tuning:
Conducted exhaustive benchmarking of 30+ Or-Tools configurations. Identified that basic "Savings" heuristics got stuck in local minima, leading to massive inefficiencies.

Phase 3 – The Breakthrough:
Discovered the optimal mathematical configuration: pairing the "Christofides" graph theory heuristic with a "Simulated Annealing" metaheuristic, reducing required fleet size significantly.

Phase 4 – Production Security:
Overcame strict browser CORS policies and Next.js caching errors by engineering a reverse-proxy tunnel using DuckDNS and Let's Encrypt SSL certificates.

Methodology

Phase 1: Digitization: Converting Stops addresses into precise GPS coordinates.

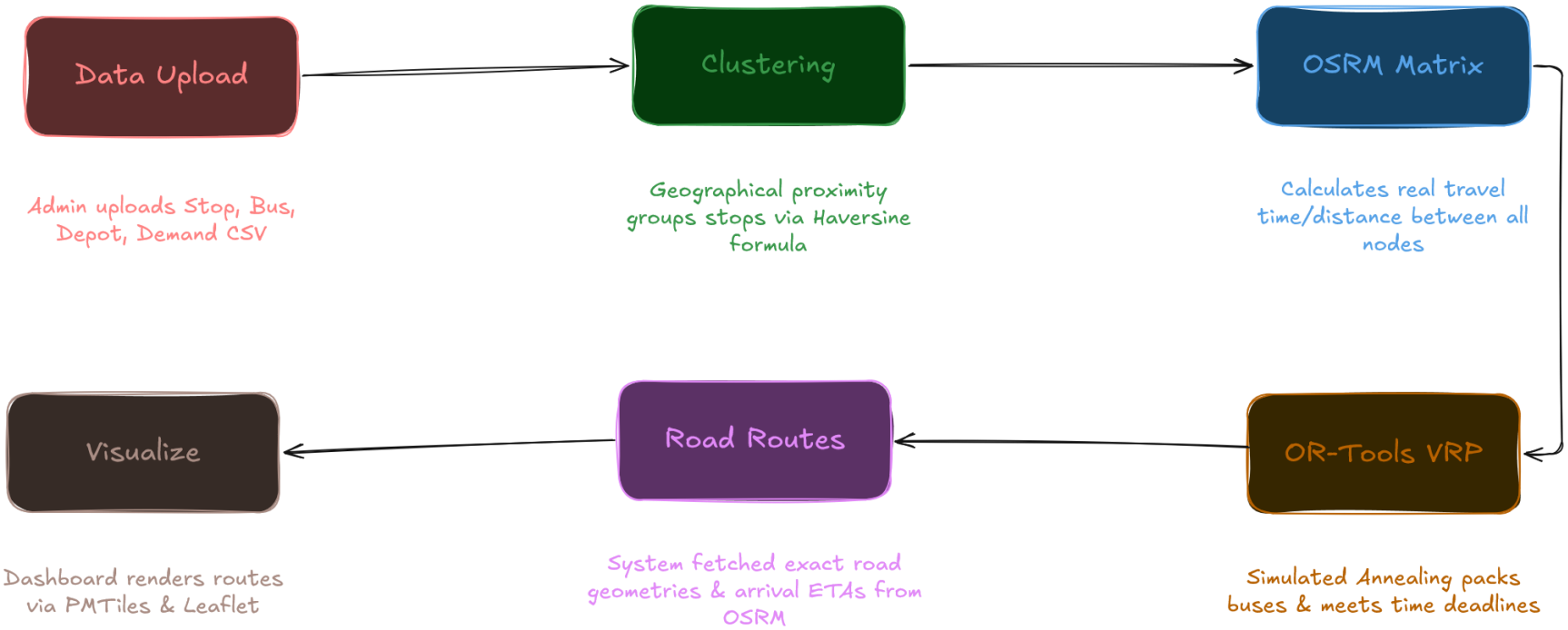
Phase 2: Matrix Generation: Using self-hosted OSRM to calculate a massive "Cost Matrix" (Time/Distance) connecting all stops, depots and the campus.

Phase 3: Optimization: Applying the “Christofides" heuristic for initial route generation, followed by “Simulated Annealing” local search to mathematically pack buses and minimizing distance.

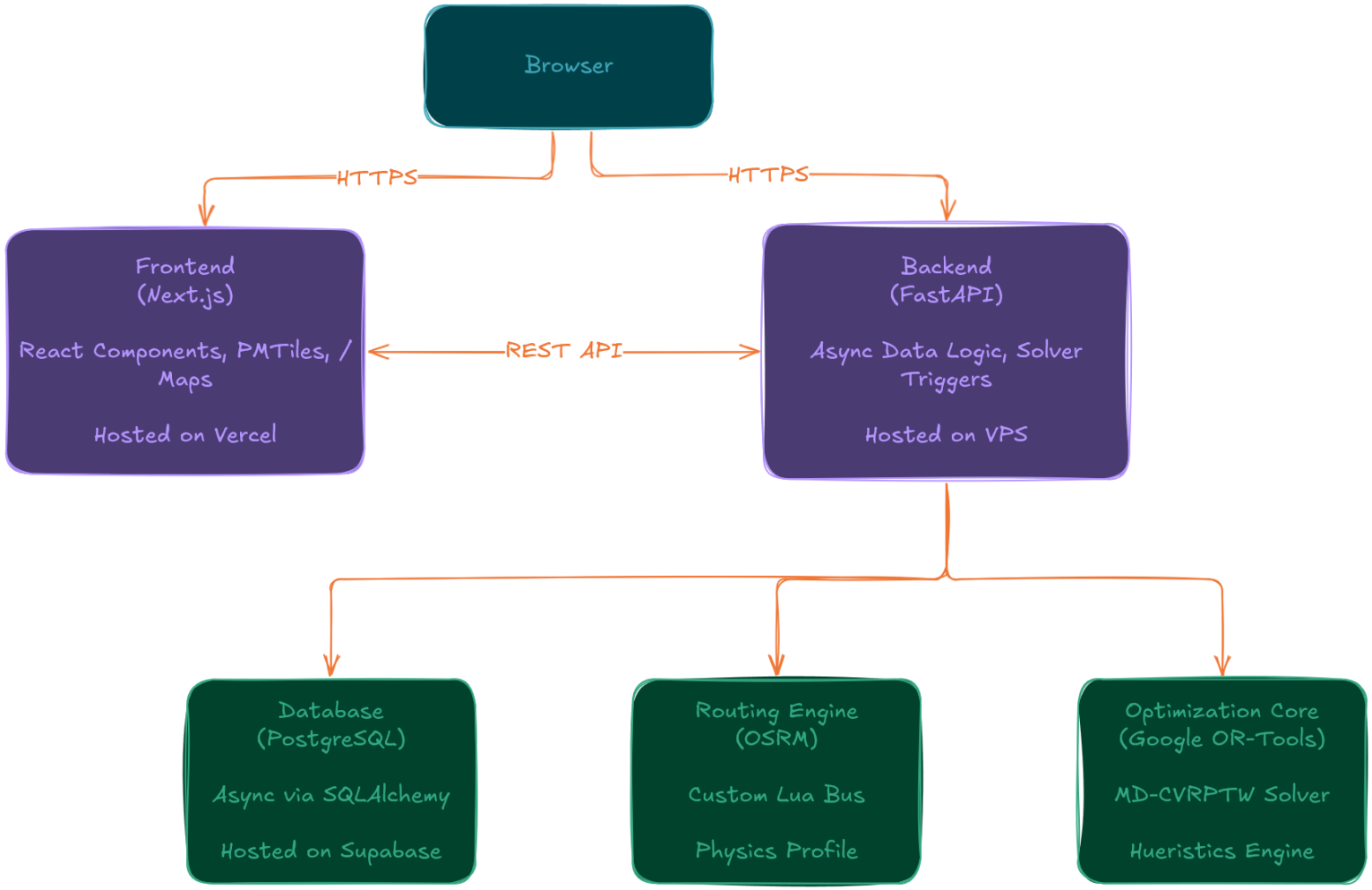
Phase 4: Visualization: Rendering the final paths on the admin dashboard using edge-stored PMTiles.

Technologies Used:
Docker: For hosting the routing engine.
PostgreSQL: For structured data storage.
Next.js: For frontend
FastAPI: For backend

Project Flow diagram



Technical Architecture



Project Timeline

Phase 1
Foundation & Data Ingestion
(Weeks 1-2)

- Setup PostgreSQL, Next.js, FastAPI
- Built CSV parsing and geographic clustering (Haversine) logic

Phase 2
Engine Integration & Physics
(Weeks 3-4)

- Self hosted OSRM in Docker using custom OpenStreetMap .pbf extracts
- Engineered custom Lua profiles to enforce heavy-bus driving physics and U-Turn penalties

Phase 3
Algorithmic Crucible
(Weeks 5-9)

- Benchmarked 30+ OR-Tools configurations
- Discarded flawed 'Savings' hueristics, implemented 'Christofides' & 'Simulated Annealing' for peak optimization

Phase 4
Polishing & Deployement
(Weeks 10-11)

- Implemented PMTiles for crash-free, high-performance map rendering
- Configured Github Actions CI/CD and resolved CORS/SSL via Nginx reverse proxy

Data & Resources

Data Sources

Map Data: OpenStreetMap (OSM) extracts for North India (via Geofabrik).
Stop Data: Geotagged coordinates for student pickup points across Palwal, Faridabad, Noida, etc.

Software & Frameworks

Frontend: Next.js (React), Leaflet.js, TanStack Query, PMTiles.
Backend: Python, FastAPI, Pandas, SQLAlchemy.
Optimization: Google OR-Tools, OSRM (Open Source Routing Machine) with custom Lua profiles.
DevOps & Infrastructure: Docker, Github Actions, Nginx Reverse Proxy, PostgreSQL.

Project Usecases & Scope

Primary Use cases:

Semester Planning:
Generating the master route schedule at the start of every academic session.

Budget Optimization:
Estimating and reducing fuel budgets by optimizing route kilometers.

Feasibility Testing:
Checking if new routes to distant towns (e.g., Rewari) are viable within time limits.

Scope:

Geographic:
Covers the entire K.R. Mangalam University catchment area (Sohna, Gurugram, Faridabad, Noida, Palwal, Bhiwadi).

Operational:
Focuses on static planning and fleet scheduling (Does not include real-time GPS tracking hardware).

Expected Results & Impact

Operational Savings:
100% elimination of software licensing costs.
Estimated 10-15% reduction in fuel consumption due to optimized paths while maintaining 100% coverage.

Efficiency:
Reduction in manual planning time from weeks to minutes.
Balanced passenger loads via dynamic split-delivery logic, preventing overcrowding on popular routes.

Sustainability:
Reduction in carbon footprint by minimizing deadhead (empty) kilometers and eliminating unnecessary U-turns and backtracking.

THANK YOU